



Pablo MARTÍNEZ-MIRAVÉ

CURRICULUM VITAE

CONTACT INFORMATION

Email: pablo.mirave@nbi.ku.dk

Webpage: <https://pamarmi.github.io>

CURRENT POSITION & EDUCATION

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|-------------------------------|--|
| October 2023 - present | Postdoctoral Fellow at Niels Bohr Institute (NBI)
University of Copenhagen |
| October 2019 - September 2023 | PhD Candidate at University of Valencia
FPU Fellowship for PhD studies (Ministry of University)
Thesis title: Neutrino properties from the laboratory and the cosmos <ul style="list-style-type: none">- Graded <u>excellent cum laude</u> and International Mention- <u>2023-2024 PhD Award from the Royal Academy of Doctors of Spain (RADE)</u> |
| October 2018 - October 2019 | Master in Advanced Physics (Specialised in Theoretical Physics)
University of Valencia
Fellowship for Introduction to Research at IFIC (9 months) <ul style="list-style-type: none">- <u>Extraordinary award</u> of the master |
| September 2014 - July 2018 | Bachelor studies in Physics
University of Valencia <ul style="list-style-type: none">- <u>Extraordinary award</u> of the bachelor |

PUBLICATIONS

In the field of particle astrophysics

Authors' order is set according to their relative contribution, following common practices in the field

- **Unveiling Neutrino Nature with the Diffuse Supernova Background**
Marco Manno, Pablo Martínez-Miravé, Irene Tamborra
e-Print: 2608.14785 [astro-ph.HE]

- **Matter- and magnetically-driven flavor conversion of neutrinos in magnetorotational collapses**
Marco Manno, Pablo Martínez-Miravé, Irene Tamborra
Class.Quant.Grav. (2026) 11, 115001 • e-Print: 2603.08843 [astro-ph.HE]
- **Detection horizon for the neutrino burst from the stellar helium flash**
Pablo Martínez-Miravé, Irene Tamborra, Georg Raffelt
Phys.Rev.D 113 (2026) 12, 123020 • e-Print: 2602.15119 [astro-ph.SR]
- **Neutrinos from stars in the Milky Way**
Pablo Martínez-Miravé, Irene Tamborra
Phys.Rev.D 113 (2026) 2, 023014 • e-Print: 2510.07399 [astro-ph.SR]
Selected Editor's Suggestion
- **Identifying Thorne–Żytkow Objects through Neutrinos**
Pablo Martínez-Miravé, Irene Tamborra, Alejandro Vigna-Gómez
Astrophys.J.Lett. 984 (2025) 1, L2 • e-Print: 2501.03330 [astro-ph.HE]
- **The Unknowns of the Diffuse Supernova Neutrino Background Hinder New Physics Searches**
M. MacDonald, P. Martínez-Miravé, I. Tamborra
JCAP 01 (2025) 062 • e-Print: 2409.16367 [astro-ph.HE]
- **Diffuse Neutrino Background from Magnetorotational Stellar Core Collapses**
P. Martínez-Miravé, I. Tamborra, M. Á. Aloy, M. Obergaulinger
Phys.Rev.D 110 (2024) 10, 103029 • e-Print: 2409.09126 [astro-ph.HE]
- **The next galactic supernova can uncover mass and couplings of particles decaying to neutrinos**
B. Telalovic, D. F. G. Fiorillo, P. Martínez-Miravé, E. Vitagliano, M. Bustamante
JCAP 11 (2024) 011 • e-Print: 2406.15506 [hep-ph]
- **The Sun and core-collapse supernovae are leading probes of the neutrino lifetime**
P. Martínez-Miravé, I. Tamborra, M. Tórtola
JCAP 05 (2024) 002 • e-Print: 2402.00116 [astro-ph.HE]

In the field of **particle physics**

Authors listed in alphabetical order, following common practices in the field.

- **The Highest-Energy Neutrino Event Constrains Dark Matter-Neutrino Interactions**
Toni Bertólez-Martínez, Gonzalo Herrera, Pablo Martínez-Miravé, Jorge Terol Calvo
e-Print: 2506.08993 [hep-ph]
- **Effects of Neutrino-Ultralight Dark Matter Interaction on the Cosmic Neutrino Background**
P. Martínez-Miravé, Y. F. Perez-Gonzalez, M. Sen
Phys.Rev.D 110 (2024) 5, 055005 • e-Print: 2406.01682 [hep-ph]
- **Neutrino CPT violation in the solar sector**
G. Barenboim, P. Martínez-Miravé, C. A. Ternes, M. Tórtola
Phys.Rev.D 108 (2023) 3, 035039 • e-Print: 2305.06384 [hep-ph]
- **Non-unitary three-neutrino mixing in the early Universe**
S. Gariazzo, P. Martínez-Miravé, O. Mena, S. Pastor, M. Tórtola
JCAP 03 (2023) 046 • e-Print: 2211.10522 [hep-ph]

- **Solar electron antineutrino flux: Revisiting bounds on neutrino magnetic moments and solar magnetic field**
E. Akhmedov, [P. Martínez-Miravé](#)
JHEP 10 (2022) 144 • e-Print: 2207.04516 [hep-ph]
- **Cosmology-friendly time-varying neutrino masses through the sterile portal**
G.-Y. Huang, M. Lindner, [P. Martínez-Miravé](#), M. Sen
Phys.Rev.D 106 (2022) 3, 033004 • e-Print: 2205.08431 [hep-ph]
- **Nonstandard interactions from the future neutrino solar sector**
[P. Martínez-Miravé](#), S. Molina Sedgwick, M. Tórtola
Phys.Rev.D 105 (2022) 3, 035004 • e-Print: 2111.03031 [hep-ph]
- **Signatures of primordial black hole dark matter at DUNE and THEIA**
V. De Romeri, [P. Martínez-Miravé](#), M. Tórtola
JCAP 10 (2021) 051 • e-Print: 2106.05013 [hep-ph]
- **Cosmological radiation density with non-standard neutrino-electron interactions**
P. F. de Salas, S. Gariazzo, [P. Martínez-Miravé](#), S. Pastor, M. Tórtola
Phys.Lett.B 820 (2021) 136508 • e-Print: 2105.08168 [hep-ph]
- **Signatures of ultralight dark matter in neutrino oscillation experiments**
A. Dev, P. A. N. Machado, [P. Martínez-Miravé](#)
JHEP 01 (2021) 094 • e-Print: 2007.03590 [hep-ph]
- **2020 global reassessment of the neutrino oscillation picture**
P. F. de Salas, D. V. Forero, S. Gariazzo, [P. Martínez-Miravé](#), O. Mena, C. A. Ternes, M. Tórtola, J. W. F. Valle
JHEP 02 (2021) 071 • e-Print: 2006.11237 [hep-ph]
- **Sterile neutrinos with altered dispersion relations revisited**
G. Barenboim, [P. Martínez-Miravé](#), C. A. Ternes, M. Tórtola
JHEP 03 (2020) 070 • e-Print: 1911.02329 [hep-ph]

Selected publications as part of Collaborations:

- **Quantum gravity phenomenology at the dawn of the multi-messenger era - A review**
A. Addazi et al. • Prog.Part.Nucl.Phys. 125 (2022) 103948 • e-Print: 2111.05659 [hep-ph]
Active contribution to several subsections on the section on Phenomenology of quantum gravity.
- **Synergy between cosmological and laboratory searches in neutrino physics: a white paper**
K. Abazajian et al. • Physics of the Dark Universe (2023) 101333 • e-Print: 2203.07377 [hep-ph]
Active contribution contributed to the section on Laboratory probes of neutrino properties.

SEMINARS, WORKSHOPS, AND CONFERENCES

- **Talk** at APCTP - NBIA Global Cooperation Workshop • 8 December 2025, Niels Bohr Institute, Copenhagen, (Denmark)
- **Invited Seminar** • 5 December 2025, DESY, Zeuthen, (Germany)
- **Parallel talk** at XXI International Workshop on Neutrino Telescopes • 29 September - 3 October 2025, Padova, (Italy)

- **Invited talk** at 5th EuCAPT Annual Symposium • 5 - 7 May 2025, CERN, (Switzerland) Online
- **Invited seminar** • 9 April 2025, INFN Sezione di Pisa, Pisa (Italy)
- **Invited seminar** • 17 March 2025, Niels Bohr Institute, Copenhagen (Denmark)
- **Talk** at MITP Topical Workshop - Towards the Detection of Diffuse Supernova Neutrinos: What will we see? What can we learn? • 16-20 September 2024, Mainz Institute for Theoretical Physics, Johannes Gutenberg University Mainz, Mainz (Germany)
- **Invited colloquium** at Physics Department at Universidad Católica del Norte • 12 September 2024, online - Physics Department at Universidad Católica del Norte, Antofagasta (Chile)
- **Invited talk** at YOUNGST@RS - Interacting dark sectors in astrophysics, cosmology, and the lab • 6 - 9 November, 2023, online - Mainz Institute for Theoretical Physics, Johannes Gutenberg University
- **Two parallel talks** at 18th International Conference on Topics in Astroparticle and Underground Physics (TAUP 2021) • 28 August - 1 September, 2023, Vienna (Austria)
- **Plenary talk** at Fourth Annual conference “COST - Quantum Gravity Phenomenology in the Multimessenger Approach” • 10 - 14 July, 2022, Rijeka (Croatia)
- **Talk** at CERN Neutrino Platform Pheno Week 2023 • 13 - 17 March, 2023, CERN (Switzerland)
- **Invited Journal Club/Seminar** • 21 November 2022, online, Technische Universität München, Munich (Germany)
- **Invited NBIA Astroparticle seminar** • 07 November 2022, Niels Bohr Institute, Copenhagen (Denmark)
- **Invited seminar** • 22 September 2022, CFTP, Instituto Superior Técnico, Lisbon (Portugal)
- **Plenary talk** at Third Annual conference “COST - Quantum Gravity Phenomenology in the Multimessenger Approach” • 12 - 15 July, 2022, Naples (Italy)
- **Parallel talk** at 9th Workshop on Flavour Symmetries and Consequences in Accelerators and Cosmology (FLASY 2022) • 27 June - 01 July, 2022, Lisbon (Portugal)
- **Invited talk** for IFIC Scientific Days of the Research line: Neutrinos and Lepton Flavour • 16-17 June, 2022, Valencia (Spain)
- **Poster and video** at XXX International Conference on Neutrino Physics and Astrophysics (Neutrino 2022) • 30 May - 4 June 2022, Online
- **Talk** at 19th MultiDark Consolider Workshop • 23 - 25 May, 2022, Miraflores de la Sierra (Spain)
- **Parallel talk** at @FlipPhysic • 21-25 March, 2022, Valencia (Spain)
- **Invited talk** for the Journal Club of INFN, Sezione di Torino • 25 February, 2022, Online
- **Invited Gravitículas Seminar**, Facultad de Física, Pontificia Universidad Católica de Chile • 25th November, online
- **Plenary (online) talk** at Second Annual conference “COST - Quantum Gravity Phenomenology in the Multimessenger Approach” • 8 - 10 March 2021, Corfu, Greece (Online participation)
- **Parallel prerecorded talk** at 17th International Conference on Topics in Astroparticle and Underground Physics (TAUP 2021) • 26 August - 6 September, 2021, Online
- **Award-winning poster** at The 28th International workshop on Weak Interactions and Neutrinos (WIN2021) • 7 - 12 June, 2021, online
- **Flash talk** at First EuCAPT Annual Symposium • 5 - 7 May, 2021, online
- **Plenary talk** at 17th MultiDark Consolider Workshop • 25 - 27 January, 2021, online
- **Poster and video** at XXIX International Conference on Neutrino Physics and Astrophysics (Neutrino 2020) • 22 June - 2 July 2020, online
- **Plenary talk** at First Annual conference “COST - Quantum Gravity Phenomenology in the Multimessenger Approach” • 10 - 13 March 2020, Granada, Spain

- **Talk** at 16th MultiDark Consolider Workshop • 25 - 27 September, 2019, La Rábida, Huelva (Spain)

TEACHING AND MENTORING

- **Invited lecturer** (4h) to the course of Fundamentals of High-Energy Astrophysics and Particle Astrophysics - Master degree in Physics, University of Copenhagen (2024-2025)
- **Teaching assistant** in Mechanics II (Theoretical classical mechanics) - Bachelor studies in physics, University of Valencia (2021 -2023)
- **Teaching assistant** in General Physics I - Bachelor studies in Chemistry, University of Valencia (2020-2021)
- **Teaching assistant** in Physics I - Bachelor studies in Mathematics and Physics, University of Valencia (2020-2021)
- Informal **supervision of MSc student** Julien Combel (University of Copenhagen) at the Niels Bohr Institute, 2024-2025.
- **Mentoring of visiting PhD student** Marco Manno (Università del Salento) at the Niels Bohr Institute, October 2024.
- **Mentoring of visiting bachelor student** Miller MacDonald (Harvard University) at the Niels Bohr Institute, June – August 2024.
- **Co-supervision of bachelor student** Lluís Mira Castelló (2022-2023)
- **Co-supervision of bachelor student** Víctor Viciach Gallego (2022-2023)
- **Co-supervision of bachelor student** Gracia Garcia
- Presenter of the **activities in Particle Astrophysics** at the NBIA **Master students' Day** (2024 and 2025)

DISEMINATION

Interviews and appearances in press

- **How Many Ghost Particles All the Milky Way's Stars Send Towards Earth** - UCPH Communication (2026)
- **Earth is bathed in droves of neutrinos spewed by the Milky Way's stars** - Science News Magazine (2026)
- **Spøgelsespartikler kan røbe mystiske hybridstjerner** (Ghost particles can reveal mysterious hybrid stars) - Videnskab.dk

Dissemination article

- **Neutrinos: Las enigmáticas partículas que han dado forma al universo** (Neutrinos: the puzzling particles which shaped the universe)
Anales de la Real Academia de Doctores de España. Volumen 10, número 3 (2025)

OUTREACH AND DIVERSITY ACTIVITIES

- **Participant at the Science Slam Copenhagen**, 23 May 2025, KU Festival, Copenhagen (Denmark)
- **High-school talks** in the context of the **International day of Women and Girls in Science** (Female pioneers in the Universe - Pioneras en el Universo).
- **High-school outreach talks on Astroparticle Physics** (Beyond light: Particle experiments in unexpected locations - Más allá de la luz: Experimentos de partículas en lugares insólitos).
- Member of the **Outreach and Diversity Committee** of COST ACTION CA18108, “Quantum Gravity phenomenology in the multi-messenger approach”. Highlighted activities include active involvement in the campaigns for the International day of Women and Girls in Science, and shared administration and content creation of the Instagram Account.
- Mentoring and supervision of high-school students during the **Masterclass Hands on Particle Physics: MINERvA** (31 March 2022, Valencia)
- **Outreach activity** “Cocinando en el Ártico” (Cooking in the Arctic) at **Open Days of Parc Científic**, XIV EXPOCIENCIA (28 May 2022, Valencia)

PEER REVIEW AND EVALUATIONS

Peer reviewer for the **Journal of High Energy Physics**, **Physics Letters B**, **Physical Review Letters**, **Physical Review D**, and **Nuclear Physics B**.

Expert (evaluator) for **Marie Skłodowska-Curie Actions Postdoctoral Fellowships 2024** Call.

LANGUAGES AND IT SKILLS

Spanish: Mother tongue

Catalan: Fluent (C1 certified - Junta Qualificadora de Coneixements de València)

English: Fluent (C1 certified - Cambridge)

French: Fluent (C1 certified - Escuela Oficial de Idiomas Valencia)

Coding languages: C/C++, Julia, Python, basic Mathematica, basic HTML5.

Specify software and packages: GLoBES, CLASS, MontePython, BlackHawk.

Miscellaneous: Microsoft Office, LaTeX, Adobe Illustrator, Adobe InDesign, Inkscape.

MANAGEMENT AND OTHER MERITS

Board Member of the Niels Bohr Institute PhDs and Postdocs Network • Copenhagen, 2024 - present

Co-organiser of the Astroparticle Journal Club at the Niels Bohr Institute • Copenhagen, 2024 - present

Local Organising Committee of the First EuCAPT School: Cosmology • Valencia (Spain), 18-22 September 2023.

Scientific secretary for TAUP 2021 and ISAPP 2021 Valencia

Active participation in the preparation of the virtual poster sessions held with the online platform GatherTown and the summary webpages with the poster display.